

Divisor-Poset Content Lifts of Fixed-Shift Primitive Entry Kernels: Zeta Inversion, Diagonal Filtrations, and Provenance-Compatible Duality

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

TPC-74 resolved every direct off-diagonal missing-native Gram entry into literal primitive kernels $L_{mn,q}(u, v)$, where the content q divides the fixed-shift difference $\Delta_{mn} = h_0(n - m)$. TPC-75 then made the cost of an arbitrary nonprimitive completion exactly primal-dual. The unrestricted optimum, however, need not preserve any q -slice provenance and therefore does not yet identify a physical construction rule.

This paper develops a provenance-compatible completion class for this TPC chain. For a fixed nonzero Δ , define the divisor-poset content lift

$$F^Z(du, dv) = \sum_{d|q||\Delta|} L_q(u, v), \quad (u, v) = 1.$$

At $d = 1$ it equals the exact post- q -aggregation primitive datum, so it is a legal analytic completion. Its diagonal profiles are the zeta transform on the divisor lattice. We prove the exact inversion

$$L_q(u, v) = \sum_{q|d||\Delta|} \mu(d/q) F^Z(du, dv),$$

so the lift retains, rather than smooths away, the complete content decomposition.

The same formula gives an exact diagonal filtration. If $(x, y) = r$ and $x = ru, y = rv$, then

$$(F^Z)^{[a]}(x, y) = \sum_{ar|q||\Delta|} L_q(u, v).$$

Consequently every auxiliary slice with $a \nmid |\Delta|$ vanishes. The high-diagonal tail is therefore supported only on actual divisors of the fixed row-pair difference, although no smallness of its mass is asserted.

We next introduce all provenance filters

$$F^W(du, dv) = \sum_{d|q||\Delta|} w_{d,q} L_q(u, v), \quad w_{1,q} = 1.$$

This affine class contains the primitive zero completion and the zeta lift. Its optimal full divisor-variation cost is attained and has the reduced exact dual

$$\min_{\theta} \|A_D(F^0 + B_L\theta)\|_1 = \max_{\substack{\|z\|_{\infty} \leq 1 \\ B_L^* A_D^* z = 0}} \operatorname{Re}\langle z, A_D F^0 \rangle.$$

Thus a provenance-compatible fill and a content-flow obstruction are again exact alternatives.

The construction is attached to the complete literal fixed- h_0 carrier with its original raw normalization. It proves neither that the zeta lift improves the zero completion nor that the optimal provenance matrix has small Perron radius. No fixed-2, parity-breaking, prime-pair, or twin-prime conclusion is claimed.

Keywords: divisor poset; zeta transform; Möbius inversion; analytic completion; content slices; diagonal filtration; convex duality; fixed shift.

Convention. Every physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete native output keys, the actual active support, and raw atomic normalization. No statement is specialized to $h_0 = 2$.

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1 Introduction

The completion problem now has an exact analytic optimum, but a physical rule must remember where its coefficients came from. For one actual row pair, TPC-74 [3] gives

$$L(u, v) = \sum_{q|\Delta} L_q(u, v), \quad \Delta = h_0(n - m) \neq 0,$$

on the primitive cofactor plane. The q -slice is not an auxiliary Fourier label: it is the greatest common content of the two literal residual factors, and its divisor restriction follows from the exact fixed-shift Diophantine identity. Aggregation over q must occur before an absolute-value estimate, because distinct slices can cancel at the same primitive coordinate.

TPC-75 [4] proves that, for a finite rectangle and full divisor-variation analysis operator A_D ,

$$\gamma_D(L) = \min_{RF=L} \|A_DF\|_1$$

has an exact strong dual. That theorem deliberately allows arbitrary nonprimitive values. Such freedom provides the sharpest separated analytic benchmark, but it may mix literal contents in a way no uniform physical rule can reproduce.

The present paper inserts an intermediate class:

$$\text{zero completion} \subset \text{content-filter completions} \subset \text{all analytic completions}.$$

The class is generated only from the actual q -resolved kernels. It therefore preserves fixed- h_0 provenance without declaring its nonprimitive values to be new atoms.

1.1 The divisor-poset lift

Every lattice point has a unique representation

$$(x, y) = (du, dv), \quad d = (x, y), \quad (u, v) = 1.$$

We fill that point with the cumulative content above d :

$$F^Z(du, dv) = \sum_{d|q|\Delta} L_q(u, v).$$

This is the upper zeta transform on the finite divisor poset of $|\Delta|$. Poset Möbius inversion [1] recovers every literal slice. Hence the lift is information-preserving on the full divisor profile even though its primitive trace sees only the aggregate.

The same cumulative structure identifies every diagonal slice exactly. Only $a \mid |\Delta|$ can contribute to the a -th divisor lift. This is a genuine carrier reduction relative to an arbitrary ambient completion. It is not a tail estimate: a function supported on few diagonal indices can still have large coefficients.

1.2 A provenance-compatible optimization

The zeta coefficient 1 is only one rule. Allow weights $w_{d,q}$ whenever $d \mid q$, while fixing $w_{1,q} = 1$ so the primitive datum is unchanged. Both the zero completion and zeta lift belong to this affine family. For fixed literal slices, the completion is linear in the free weights, so its exact TPC-75 cost becomes a smaller finite convex program. We derive its reduced strong dual. The dual stationarity condition says that the analysis charge annihilates every permitted content-flow direction.

1.3 What is and is not gained

The new results are exact:

- the lift, inverse, diagonal support, and Hermitian reversal;
- a sharp support census in terms of the actual q -slice supports;
- the full provenance-filter hierarchy;
- primal and dual attainment in that hierarchy; and
- exact operator upper and obstruction matrices after row-pair aggregation.

They do not order the zero and zeta costs universally. Lower $d = 1$ variation can be exchanged for higher diagonal mass. They also do not turn the divisor bound $\#\{d : d \mid \Delta\} = \tau(|\Delta|)$ into a normalized saving.

1.4 Organization

Section 2 proves the zeta lift and inversion. Section 3 derives the exact diagonal hierarchy and support census. Section 4 introduces all content filters and their reduced dual. Section 5 handles reversal and full matrices. Section 6 records the non-substitution laws. Section 7 states the actual fixed-shift gate, and section 8 gives the audit ledger.

2 The divisor-poset content lift

2.1 Literal content data

Fix an integer $\Delta \neq 0$ and put

$$\mathcal{D}_\Delta = \{q \in \mathbb{N} : q \mid |\Delta|\}.$$

Let P be a finite set of primitive pairs. For every $q \in \mathcal{D}_\Delta$, let

$$L_q : P \longrightarrow \mathbb{C}$$

be a finitely supported content slice, extended by zero to all primitive pairs. Its physical aggregate is

$$k(u, v) = \sum_{q \in \mathcal{D}_\Delta} L_q(u, v).$$

This abstract notation will later be specialized to $L_{mn,q}$, with all complete native key labels already summed inside the literal fiber.

Choose a finite rectangle B containing every point (du, dv) for which $d \mid q \mid |\Delta|$ and $(u, v) \in \text{supp } L_q$. Values outside B are zero.

Definition 2.1 (Content zeta lift). For $x, y \geq 1$, write

$$d = (x, y), \quad x = du, \quad y = dv.$$

Define

$$(\mathfrak{Z}_\Delta \mathbf{L})(x, y) = \mathbf{1}_{d \mid |\Delta|} \sum_{\substack{q \mid |\Delta| \\ d \mid q}} L_q(u, v). \tag{1}$$

We denote this kernel by $F^\mathbb{Z}$.

Proposition 2.2 (Completion and provenance). *The zeta lift is a legal completion of k :*

$$F^\mathbb{Z}(u, v) = k(u, v) \quad ((u, v) = 1).$$

Every nonzero nonprimitive value uses only literal slices whose actual content is a multiple of the inserted common factor.

Proof. At a primitive site $d = 1$, every $q \in \mathcal{D}_\Delta$ occurs in (1), giving the aggregate k . At a nonprimitive site, the displayed divisibility condition is exactly $d \mid q$. $\square$

2.2 Exact zeta inversion

For every divisor $d \mid |\Delta|$, define the diagonal profile

$$G_d(u, v) = F^Z(du, dv) \quad ((u, v) = 1).$$

Then

$$G_d(u, v) = \sum_{\substack{q \mid |\Delta| \\ d \mid q}} L_q(u, v). \quad (2)$$

For notational convenience in formulas carrying a divisibility indicator, extend this definition by $G_e = 0$ whenever $e \nmid |\Delta|$.

Theorem 2.3 (Divisor-poset inversion). *For every $q \mid |\Delta|$ and primitive (u, v) ,*

$$L_q(u, v) = \sum_{\substack{d \mid |\Delta| \\ q \mid d}} \mu(d/q) G_d(u, v). \quad (3)$$

Thus the full profile $(G_d)_{d \mid |\Delta|}$ and the literal slice family $(L_q)_{q \mid |\Delta|}$ determine each other uniquely.

Proof. For a fixed primitive coordinate, (2) is the upper zeta transform on the divisor poset. Its incidence-algebra inverse is

$$\mu_{\mathcal{D}_\Delta}(q, d) = \mu(d/q) \quad (q \mid d),$$

the ordinary number-theoretic Möbius function on the quotient interval. Apply finite Möbius inversion. $\square$

Remark 2.4 (The primitive trace is not invertible). The primitive datum is only $G_1 = \sum_q L_q$. It cannot recover the individual slices. Inversion requires the entire nonprimitive divisor profile. Thus the zeta lift uses literal provenance that is additional to the aggregated completion datum, while preserving the same physical functional.

Corollary 2.5 (Cancellation can reappear below the primitive layer). *If $|\Delta| > 1$, it is possible that $G_1 = 0$ but $G_d \neq 0$ for some $d > 1$. Consequently the zeta lift of a zero aggregate need not be the zero completion.*

Proof. Choose any divisor $r > 1$ of $|\Delta|$, take $L_1 = F, L_r = -F$, and set the other slices to zero. Then $G_1 = 0$, while $G_r = L_r = -F$. $\square$

3 Exact diagonal filtration

3.1 Nested profiles

For a finitely supported kernel F , recall

$$F^{[a]}(x, y) = F(ax, ay).$$

Write

$$r = (x, y), \quad x = ru, \quad y = rv, \quad (u, v) = 1.$$

Theorem 3.1 (Diagonal filtration identity). *For the content zeta lift and every $a, x, y \geq 1$,*

$$(F^Z)^{[a]}(x, y) = \mathbf{1}_{ar \mid |\Delta|} G_{ar}(u, v) = \mathbf{1}_{ar \mid |\Delta|} \sum_{\substack{q \mid |\Delta| \\ ar \mid q}} L_q(u, v). \quad (4)$$

In particular,

$$(F^Z)^{[a]} \equiv 0 \quad \text{whenever } a \nmid |\Delta|. \quad (5)$$

Proof. The greatest common divisor of ax and ay is ar , and their primitive reduction is (u, v) . Apply equations (1) and (2). If $a \nmid |\Delta|$, then $ar \nmid |\Delta|$ for every r , so the slice vanishes. $\square$

Remark 3.2 (A filtration, not independent copies). As the diagonal index grows through divisors of $|\Delta|$, the allowed content set shrinks:

$$a_1 \mid a_2 \implies \{q : a_2 \mid q \mid |\Delta|\} \subseteq \{q : a_1 \mid q \mid |\Delta|\}.$$

The higher slices are nested content tails. Treating them as unrelated ambient kernels discards this exact dependence.

3.2 Exact tail restriction

The TPC-71 exact diagonal tail [2] is

$$\mathcal{T}_D(F) = \sum_{a > D} |\mu(a)| \sum_{x, y \geq 1} |\mu(ax)\mu(ay)F(ax, ay)|.$$

Corollary 3.3 (Divisor-supported tail). *For the zeta lift,*

$$\boxed{\mathcal{T}_D(F^{\mathbb{Z}}) = \sum_{\substack{a \mid |\Delta| \\ a > D}} |\mu(a)| \sum_{x, y \geq 1} \left| \mu(ax)\mu(ay) \mathbf{1}_{a(x, y) \mid |\Delta|} G_{a(x, y)} \left(\frac{x}{(x, y)}, \frac{y}{(x, y)} \right) \right|}. \quad (6)}$$

Every tail index not dividing $|\Delta|$ is absent exactly.

Proof. Insert (4) into the definition of $\mathcal{T}_D$, and delete the identically zero indices by (5). $\square$

Remark 3.4 (No tail saving yet). The number of possible diagonal indices is at most $\tau(|\Delta|)$, but (6) is weighted by the actual complex amplitudes and their multiplicities. A divisor-count bound does not imply smallness relative to raw atomic mass.

3.3 Actual support census

Let

$$S_q = \text{supp } L_q$$

inside the primitive plane and

$$S_{\geq d} = \bigcup_{\substack{q \mid |\Delta| \\ d \mid q}} S_q.$$

Proposition 3.5 (Layered support bound). *The content layer with greatest common divisor d satisfies*

$$\text{supp } G_d \subseteq S_{\geq d}.$$

Consequently,

$$\# \text{supp } F^{\mathbb{Z}} = \sum_{d \mid |\Delta|} \# \text{supp } G_d \leq \sum_{d \mid |\Delta|} \# S_{\geq d} \leq \tau(|\Delta|) \# \bigcup_{q \mid |\Delta|} S_q. \quad (7)$$

All inequalities may be strict because literal q -slices can cancel inside a layer.

Proof. Equation (2) gives the first inclusion. Distinct greatest common divisors define disjoint lattice layers, giving the equality. Sum the inclusions and use $S_{\geq d} \subseteq \bigcup_q S_q$. $\square$

Corollary 3.6 (All provenance filters share the divisor carrier). *Every completion of the form*

$$F(du, dv) = \sum_{\substack{q \mid |\Delta| \\ d \mid q}} c_{d, q} L_q(u, v)$$

vanishes outside the greatest-common-divisor layers $d \mid |\Delta|$, and its layer- d support is contained in $S_{\geq d}$. In particular, it satisfies the same two cardinality upper bounds as the right-hand side of (7).

Proof. Only the coefficients change; the allowed incidences do not. $\square$

4 The provenance-compatible completion gauge

4.1 Content filters

Let

$$\mathcal{I}_\Delta = \{(d, q) : d, q \mid |\Delta|, d \mid q\}.$$

Definition 4.1 (Provenance filter). A content filter is an array

$$W = (w_{d,q})_{(d,q) \in \mathcal{I}_\Delta}$$

satisfying

$$w_{1,q} = 1 \quad (q \mid |\Delta|).$$

Its completion is

$$F^W(du, dv) = \sum_{\substack{q \mid |\Delta| \\ d \mid q}} w_{d,q} L_q(u, v), \quad (u, v) = 1. \quad (8)$$

Every F^W is a completion of the same post- q -aggregation datum. Two distinguished filters are:

$$\begin{aligned} W^0 : \quad w_{1,q}^0 &= 1, & w_{d,q}^0 &= 0 \quad (d > 1), \\ W^Z : \quad w_{d,q}^Z &= 1 & (d \mid q). \end{aligned}$$

They give the primitive zero completion F^0 and the zeta lift F^Z , respectively.

Remark 4.2 (Analytic coefficients). The numbers $w_{d,q}$ are analytic copying weights. They are not physical multiplicities, new contents, or changes to the raw row mass. Complex weights are allowed in the pair-dependent analytic benchmark. A single reversal-invariant rule will later be taken real, or conjugated on the reversed edge.

4.2 Reduced linear parametrization

Let

$$\Theta_\Delta = \mathbb{C}^{\{(d,q) \in \mathcal{I}_\Delta : d > 1\}}.$$

For the fixed literal slice family $\mathbf{L} = (L_q)$, define

$$B_{\mathbf{L}} : \Theta_\Delta \longrightarrow \mathbb{C}^B$$

by writing each $(x, y) \in B$ uniquely as $(x, y) = (du, dv)$, where $d = (x, y)$ and $(u, v) = 1$, and setting

$$(B_{\mathbf{L}}\theta)(du, dv) = \mathbf{1}_{d>1} \sum_{\substack{q \mid |\Delta| \\ d \mid q}} \theta_{d,q} L_q(u, v). \quad (9)$$

Then every filter parameter gives a provenance completion through

$$F^W = F^0 + B_{\mathbf{L}}\theta,$$

and every provenance completion arises this way. The representation need not be unique: different θ can yield the same kernel when the actual slice columns are linearly dependent.

Let A_D be the full divisor-variation analysis operator of TPC-75 [4].

Definition 4.3 (Provenance gauge).

$$\gamma_D^{\text{prov}}(\mathbf{L}) = \min_{\theta \in \Theta_\Delta} \|A_D(F^0 + B_{\mathbf{L}}\theta)\|_1.$$

Theorem 4.4 (Reduced content-flow dual). *The minimum is attained and*

$$\gamma_D^{\text{prov}}(\mathbf{L}) = \max_{\substack{\|z\|_\infty \leq 1 \\ B_{\mathbf{L}}^* A_D^* z = 0}} \text{Re}\langle z, A_D F^0 \rangle. \quad (10)$$

The dual maximum is attained.

Proof. Put $b = A_D F^0$ and $C = A_D B_L$. Under the real pairing, the polar of the complex ℓ^1 -ball is the coordinatewise unit polydisc. Finite-dimensional distance-to-a-subspace duality therefore gives

$$\inf_{\theta} \|b + C\theta\|_1 = \max_{\substack{\|z\|_{\infty} \leq 1 \\ z \perp_{\mathbb{R}} \text{Ran } C}} \text{Re}\langle z, b \rangle.$$

The annihilator condition is exactly $C^* z = 0$, which is the displayed constraint because $C = A_D B_L$. The image $b + \text{Ran } C$ is a closed affine subspace, so its distance from the origin is attained. The dual feasible set is compact, so the maximum is attained as well. $\square$

Corollary 4.5 (Exact provenance obstruction). *Every z satisfying*

$$\|z\|_{\infty} \leq 1, \quad B_L^* A_D^* z = 0$$

proves that all content-filter completions have cost at least

$$\text{Re}\langle z, A_D F^0 \rangle.$$

At the optimum this lower obstruction equals the best provenance fill.

Corollary 4.6 (Completion hierarchy). *Let $\gamma_D(k)$ be the unrestricted TPC-75 completion gauge of the aggregate primitive datum. Then*

$$\boxed{|\mathcal{S}_{\text{prim}}(k)| \leq \gamma_D(k) \leq \gamma_D^{\text{prov}}(L) \leq \min\{\mathcal{C}_D(F^0), \mathcal{C}_D(F^Z)\}.} \quad (11)$$

Proof. The physical lower bound and unrestricted gauge are TPC-75. The provenance class is a subclass of all analytic completions, while W^0 and W^Z are feasible provenance filters. $\square$

Remark 4.7 (Parameter count is not rank). The number of free weights is

$$\sum_{q|\Delta|} (\tau(q) - 1).$$

The actual dimension is $\text{rank } B_L$, which can be much smaller because a literal slice may vanish or different content-flow columns may coincide. A sparse implementation should delete the kernel of B_L before optimization.

4.3 KKT balance

Proposition 4.8 (Content-flow KKT condition). *A provenance fill θ_* and dual charge z_* are jointly optimal if and only if*

$$\begin{aligned} B_L^* A_D^* z_* &= 0, \\ \|z_*\|_{\infty} &\leq 1, \\ \text{Re}\langle z_*, A_D(F^0 + B_L \theta_*) \rangle &= \|A_D(F^0 + B_L \theta_*)\|_1. \end{aligned}$$

Thus the optimal analysis charge is balanced against every allowed content-flow direction.

Proof. These are primal feasibility, dual feasibility, and equality in the support-function bound of [theorem 4.4](#). $\square$

5 Hermitian reversal and matrix aggregation

5.1 Reversed content profiles

Return to the actual row-pair notation. TPC-74 proves

$$L_{nm,q}(v, u) = \overline{L_{mn,q}(u, v)} \quad (m \neq n).$$

The divisor set of

$$|\Delta_{mn}| = |h_0(n - m)|$$

is unchanged by reversal.

Proposition 5.1 (Filtered Hermitian reversal). *If the reverse filter obeys*

$$w_{d,q}^{nm} = \overline{w_{d,q}^{mn}},$$

then

$$F_{nm}^{W^{nm}}(v, u) = \overline{F_{mn}^{W^{mn}}(u, v)}.$$

In particular, the zero and zeta lifts are Hermitian compatible, and any single real filter can be used in both orientations.

Proof. Conjugate (8), exchange u, v , and use the literal slice reversal. $\square$

Fix transpose rectangles and equal cutoffs:

$$B_{nm} = B_{mn}^T, \quad D_{nm} = D_{mn}.$$

The divisor–variation cost is invariant under transpose conjugation. Thus the pair-dependent provenance gauges satisfy

$$\gamma_{nm}^{\text{prov}} = \gamma_{mn}^{\text{prov}}.$$

5.2 The provenance matrix

Let Q^{all} be the matrix of unrestricted TPC–75 edge gauges and let

$$Q_{mn}^{\text{prov}} = \begin{cases} \gamma_{mn}^{\text{prov}}, & m \neq n, \\ 0, & m = n. \end{cases}$$

Let P^0 and P^Z contain the costs of the zero and zeta completions, respectively, with zero diagonal. More generally, for a family of filters $\mathbf{W} = (W^{mn})_{m < n}$, put

$$P(\mathbf{W})_{mn} = \mathcal{C}_{D_{mn}}(F_{mn}^{W^{mn}}) \quad (m \neq n), \quad P(\mathbf{W})_{mm} = 0,$$

and mirror the upper triangle. The superscript “all” means all post- q -aggregation analytic completions of the fixed primitive kernel; it does not mean a pre- q , slice-by-slice optimization.

Theorem 5.2 (Four-level completion hierarchy). *For the gauged direct-entry matrix $\tilde{H}$,*

$$\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^{\text{all}}) \leq \rho(Q^{\text{prov}}) \leq \min\{\rho(P^0), \rho(P^Z)\}. \quad (12)$$

Proof. Apply [equation \(11\)](#) to every unordered row pair. All matrices are symmetric and nonnegative, so entrywise order passes to Perron spectral radius. The first two inequalities are the TPC–75 absolute-entry floor. $\square$

Theorem 5.3 (Independent provenance factorization). *If every unordered row pair may choose its own provenance filter before the later matrix extremizer is selected, then*

$$\inf_{\mathbf{W}} \rho(P(\mathbf{W})) = \rho(Q^{\text{prov}}),$$

and the infimum is attained.

Proof. Each edge minimum is attained by [theorem 4.4](#). Choose all finitely many upper-triangle minimizers simultaneously and conjugate them on the reverse edges. Every other filter family gives an entrywise larger nonnegative majorant, so Perron monotonicity proves equality. $\square$

Remark 5.4 (A universal filter is a different problem). A single formula $w_{d,q}$ shared across row pairs, blocks, scales, or a finite coefficient family couples the edge variables and can be more expensive than $\rho(Q^{\text{prov}})$. It must be optimized with the fixed-data affine coupled-rule theorem of TPC–75. Uniformity over a growing coefficient class adds a minimax quantifier and is not obtained by choosing the best filter after each literal kernel is known.

5.3 Dual obstruction matrices

For every edge, a feasible charge in [equation \(10\)](#) gives

$$O_{mn}^{\text{prov}} = \left| \text{Re} \langle z_{mn}, A_{D_{mn}} F_{mn}^0 \rangle \right| \leq \gamma_{mn}^{\text{prov}}.$$

Indeed, dual feasibility is invariant under $z \mapsto -z$, so the absolute dual objective is still bounded by the optimum. Mirror this value and put zero on the diagonal.

Corollary 5.5 (Perron obstruction).

$$\rho(O^{\text{prov}}) \leq \rho(Q^{\text{prov}}).$$

Thus a certified nonnegligible obstruction matrix closes the declared provenance-filter majorant at that scale. It does not lower-bound $\|H\|$, because the latter may exploit signed cancellation below the absolute Perron floor.

Proof. Use entrywise monotonicity. □

6 Countermodels and exact comparison examples

For this section write

$$M(x) = \sum_{n \leq x} \mu(n), \quad \Delta_{12}F(x, y) = F(x, y) - F(x+1, y) - F(x, y+1) + F(x+1, y+1),$$

and retain the TPC-75 notation $\mathcal{C}_D(F) = \|A_DF\|_1$ for the complete low-plaquette plus exact-tail cost.

6.1 The zeta lift can beat the zero completion

Take $U = V = 4$, $D = 1$, and $\Delta = 12$. Put all content in the rectangle $B = [1, 4]^2$. For every primitive $(u, v) \in B$, set

$$t(u, v) = \left\lfloor \frac{4}{\max\{u, v\}} \right\rfloor, \quad G_d(u, v) = \mathbf{1}_{d|12} \mathbf{1}_{d \leq t(u, v)},$$

and define the literal slices by the exact inverse transform

$$L_q(u, v) = \sum_{q|d|12} \mu(d/q) G_d(u, v).$$

If $L_q(u, v) \neq 0$, then necessarily $q \leq t(u, v)$; hence every inserted point (du, dv) , $d \mid q$, still lies in B . Thus this is a full lift on the declared rectangle, not a truncation. By construction its zeta profiles are the displayed G_d , so the zeta lift is the constant 4-by-4 rectangle:

$$F^Z(x, y) = 1 \quad (1 \leq x, y \leq 4).$$

Its primitive aggregate is also identically 1 on the primitive sites of B .

For the zero completion, the $d = 1$ plaquette terms with nonzero cost are

(x, y)	(1, 1)	(1, 3)	(1, 4)	(3, 1)	(3, 3)	(3, 4)	(4, 1)	(4, 3)
$ M(x)M(y)\Delta_{12}F^0(x, y) $	1	1	1	1	2	1	1	1.

Its higher diagonal tail vanishes, so

$$\mathcal{C}_1(F^0) = 9.$$

For the constant zeta rectangle, only the terminal $d = 1$ plaquette contributes, with cost 1. Its exact tail has one unit contribution at $a = 2$ and one at $a = 3$. Hence

$$\mathcal{C}_1(F^Z) = 1 + 1 + 1 = 3 < 9.$$

This is an exact example in the full cost, not merely a comparison of $d = 1$ variation.

6.2 The zero completion can beat the zeta lift

Take $D = 1$, $\Delta = 2$, one primitive atom at $(1, 1)$, and

$$L_1(1, 1) = 1, \quad L_2(1, 1) = -1.$$

The aggregate primitive datum is zero, so $F^0 = 0$ and $\mathcal{C}_1(F^0) = 0$. But

$$F^{\mathbb{Z}}(2, 2) = L_2(1, 1) = -1.$$

The tail coordinate $a = 2, x = y = 1$ has unit weight, hence

$$\mathcal{C}_1(F^{\mathbb{Z}}) \geq 1.$$

Therefore neither distinguished completion universally dominates the other.

6.3 Zeta inversion is not smoothing

If $L_q(u_0, v_0) = 1$ for every $q \mid |\Delta|$ at one primitive coordinate, then

$$G_d(u_0, v_0) = \#\{q : d \mid q \mid |\Delta|\}.$$

The total absolute profile mass is

$$\sum_{d \mid |\Delta|} |G_d(u_0, v_0)| = \sum_{q \mid |\Delta|} \tau(q),$$

whereas the literal slice mass is only $\tau(|\Delta|)$. The zeta transform is an invertible provenance expansion, not a contraction.

6.4 Few diagonal indices need not mean a small tail

Take one large coefficient A at a prime content $q > D$, supported at the primitive coordinate $(1, 1)$. The zeta lift places A on the layer $d = q$. The tail coordinate $(d, x, y) = (q, 1, 1)$ then has weight

$$|\mu(q)\mu(q)\mu(q)| = 1$$

and contributes $|A|$, even though only one high diagonal index need be active. Sparsity of divisor indices and smallness of normalized mass are different statements.

6.5 Optimizing after aggregation is broader

The unrestricted gauge $\gamma_D(k)$ sees only the aggregate primitive datum $k = \sum_q L_q$. The provenance gauge also sees the decomposition (L_q) . Two slice families with the same aggregate can therefore have the same unrestricted gauge but different provenance programs, different zeta lifts, and different dual annihilators

$$B_L^* A_D^* z = 0.$$

One may not use the cheaper decomposition unless it is the literal one carried by the complete physical keys.

6.6 A fill is not a new collision

At $d > 1$, the number

$$F^W(du, dv)$$

is an analytic copy of amplitudes already present at primitive coordinates. Counting it as an additional native key would change both the Gram entry and the raw row mass. The completion invariance theorem allows the value only because the physical Möbius functional samples primitive sites.

6.7 Pairwise filters are not a universal rule

Let one shared real parameter t control two row pairs, with optimal cost profiles

$$p_{12}(t) = |t|, \quad p_{23}(t) = |1 - t|.$$

Each pair-dependent provenance optimum is zero, but a single filter pays at least $1/2$. The exact independent factorization in [theorem 5.3](#) cannot be re-labelled as a uniform filter theorem.

7 The actual fixed-shift gate

7.1 Literal specialization

Fix one nonzero h_0 , one declared external block, both direct sides through their proved orthogonal sum, and the complete TPC-74 missing-high key carrier. For every active $m < n$,

$$\Delta_{mn} = h_0(n - m) \neq 0.$$

The slices $L_{mn,q}$ retain all complete native key labels and the raw normalization

$$\frac{1}{\sqrt{W_M(m)W_M(n)}}.$$

Here $W_M(m) = W_m$ is the sum of the raw masses of all actual atoms on both orthogonal direct sides in the declared TPC-74 block. It is not recomputed by side, degree, content, or support, and analytic copies at $d > 1$ never enter W_m . They are aggregated in the physical primitive datum before any absolute value:

$$k_{mn} = \sum_{q \mid |\Delta_{mn}|} L_{mn,q}.$$

Choose B_{mn} to contain every actual primitive slice support and every declared layer $d \mid q \mid |\Delta_{mn}|$. Fix D_{mn} , with transpose rectangle and equal cutoff on the reversed pair. These choices are made before the provenance optimization.

7.2 What the divisor filtration closes

The content lift now has three exact physical compatibilities:

- (i) every fill coefficient comes from an actual $q \mid |h_0(n - m)|$ slice;
- (ii) every nonzero greatest-common-divisor layer satisfies $d \mid |h_0(n - m)|$; and
- (iii) the entire lifted profile can be inverted back to the literal q -slices.

No smooth ambient interpolation has these properties automatically. They are L1 carrier statements because the $L_{mn,q}$ are the actual complete-key kernels of TPC-74.

7.3 The remaining positive estimate

Let Q_X^{prov} be the complete matrix of optimal provenance-filter costs on a growing physical family. A positive completion result requires an estimate such as

$$\rho(Q_X^{\text{prov}}) \leq \eta_X < 1,$$

or a fixed predeclared filter family with an equally strong complete Perron bound. Then TPC-75 gives the corresponding eigenvalue control for the normalized missing-native Gram.

Neither

$$a \mid |h_0(n - m)| \quad \text{nor} \quad \tau(|h_0(n - m)|) = |h_0(n - m)|^{o(1)}$$

proves that estimate. The first is a support identity; the second is an ambient divisor count. The required quantity contains the actual complex amplitudes, raw masses, row degrees, and restricted Mertens weights.

7.4 Negative outcomes

There are three distinct stop rules.

- (G1) If a dual content-flow matrix O_X^{prov} has nonnegligible Perron radius, the entire declared provenance-filter class is too expensive at that scale.
- (G2) If the smaller actual absolute floor $\rho(|\tilde{H}_X|)$ is nonnegligible, every entrywise absolute completion architecture is blocked, including unrestricted completions.
- (G3) Neither obstruction lower-bounds the signed norm $\|H_X\|$. A signed phase, full-moment, or hybrid anchor/direct route remains logically open.

7.5 TPC-77 decision

The next step is to exploit sparsity before attempting asymptotics. For every actual row pair, form the bipartite incidence complex linking:

$$\text{free content weights} \longleftrightarrow \text{nonprimitive sites} \longleftrightarrow \text{plaquette/tail coordinates}.$$

The next paper is planned to target exact component deletion and primal/dual factorization, together with a certificate format in which every finite census value is independently checked by matching fills and charges. Solver failure will not be called an obstruction, and finite data will not be called L2 progress.

7.6 Endpoint ledger

Even a successful Q^{prov} estimate does not close terminal activation, represented-frame recovery, missing-mass comparison, nuisance shorting, endpoint cells, or final packet reconstruction. The same strict endpoint rule remains: if total fixed-power loss reaches $1/400$, including equality, the current implication stops.

8 Claim ledger and conclusion

8.1 Exact conclusions

The paper proves:

- (1) a literal divisor-poset zeta lift of every q -resolved primitive entry family;
- (2) exact Möbius inversion recovering every q -slice from the lifted diagonal profile;
- (3) the diagonal filtration $(F^{\mathbb{Z}})^{[a]} = 0$ for $a \nmid |\Delta|$;
- (4) an exact divisor-supported high-diagonal tail formula;
- (5) layered actual-support bounds that retain cancellation;
- (6) the complete affine class of provenance filters, containing both zero and zeta completions;
- (7) an attained reduced primal and dual content-flow gauge;
- (8) Hermitian reversal and independent-edge Perron factorization;
- (9) an exact hierarchy from the signed norm through unrestricted, provenance, zero, and zeta majorants; and
- (10) explicit examples proving that neither zero nor zeta completion universally dominates the other.

8.2 Statements not proved

The paper does not prove:

- that the zeta lift improves the zero completion on the actual growing carrier;
- that the provenance optimum has a mass-normalized saving;
- that divisor-supported diagonal sparsity makes the exact tail small;
- a complete row-sum or Perron decay estimate;
- a single filter uniform over a growing coefficient class;
- that an obstruction for provenance filters covers unrestricted analytic completions;
- that failure of an absolute majorant makes the signed matrix large;
- closure of any later physical reassembly gate;
- the strict $1/400$ endpoint inequality;
- a result special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

8.3 Audit levels

Result	Scope	Level
Zeta lift, inversion, diagonal filtration	All finite divisor-resolved primitive kernels	L0
Provenance gauge and reduced dual	All finite fixed-slice affine filter classes	L0
Hermitian/Perron factorization	All finite independent unordered-edge classes	L0
Literal content specialization	Complete TPC-74 fixed- h_0 missing-high carrier	L1
Normalized provenance saving	Complete growing physical row family	Open L2
Physical reassembly and endpoint	One common carrier and coefficient scope	Open

8.4 Conclusion

A q -faithful nonprimitive completion in this TPC chain is now exact. It is the divisor-poset zeta transform

$$F^Z(du, dv) = \sum_{d|q||\Delta|} L_q(u, v),$$

and its full diagonal profile is invertible. At the level of the analytic profile, the lift therefore neither loses nor ambiguously mixes the literal content data. It does not create physical atoms: it reorganizes the given amplitudes into nested common-factor layers, all supported on divisors of the same fixed row-pair difference.

The zeta rule is not privileged analytically. The broader provenance-compatible gauge optimizes every legal content-flow weight and has the exact dual condition

$$B_L^* A_D^* z = 0.$$

This turns a vague search for a good q -compatible interpolation into a matched fill/obstruction problem. The comparison examples show why both outcomes are needed: zeta can lower the complete cost, and it can also be strictly worse than the zero completion.

The next bottleneck is no longer algebraic definition but sparse actual incidence. TPC-77 is planned to investigate decomposition of that incidence into independently certifiable components. Only after a real carrier census identifies which components dominate will it be sensible to spend a number-theory gate on their asymptotic mass.

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